

Basic Principles of Medicine 1

Module : Musculoskeletal

DLA

Development of the limbs

Part 1

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Recommended Reading

The Developing Human 11th edn – Clinically Oriented Embryology

Moore K.L, Persaud T.V.N, Torchia M.G.

Chapter 16 pages 341 - 54

Review early embryology DLA lecture slides

Watch this video:

<https://premiumbasicsscience.lovehealthlibrary.com/multimediaplayer.aspx?MultimediaID=19450537>

Pay special attention to the **green boxes** which contain the clinical correlates for this topic.

Special note

Some slides have a green background, this will not be covered in the video recording but are put so you have the information available

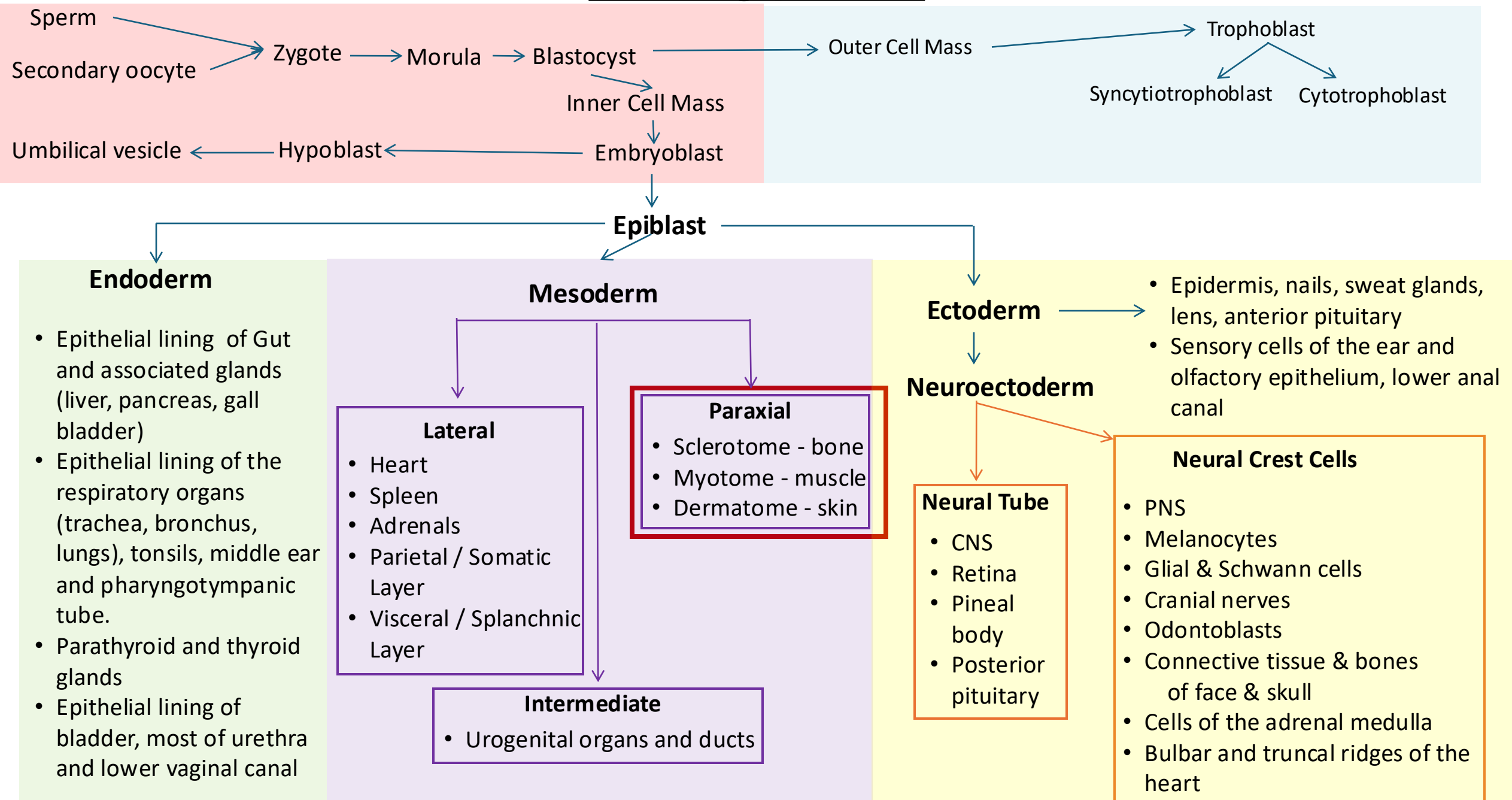
Important notes

Because of the orientation of the embryo, the terms dorsal and ventral do not have the same meaning as anterior and posterior.

Objectives

SOM.MKII.BPM1.1.MSK.1.ANAT.1343	Describe the musculoskeletal development of the limbs from the lateral intraembryonic mesoderm and myotomes of the somites.
SOM.MKII.BPM1.1.MSK.1.ANAT.1344	Asses the importance of sonic hedgehog, fibroblast growth factor and WNT in the development of the limbs
SOM.MKII.BPM1.1.MSK.1.ANAT.1345	Explain the formation of the following: congenital talipes (clubfoot), amelia / meromelia, cutaneous syndactyly, osseous syndactyly, brachydactyly, polydactyly, ectrodactyly (cleft hand or partial adactyly) and limb deformities caused by amniotic bands.
SOM.MKII.BPM1.1.MSK.1.ANAT.1346	List and explain the clinical significance of the VACTERL association

The Big picture

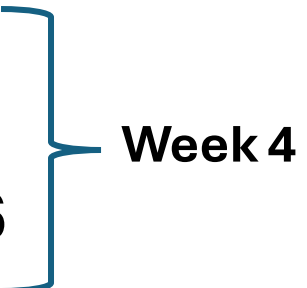


The journey of development

- Fertilization to gastrulation - Some primordial cell differentiation happens - **Weeks 1-3**

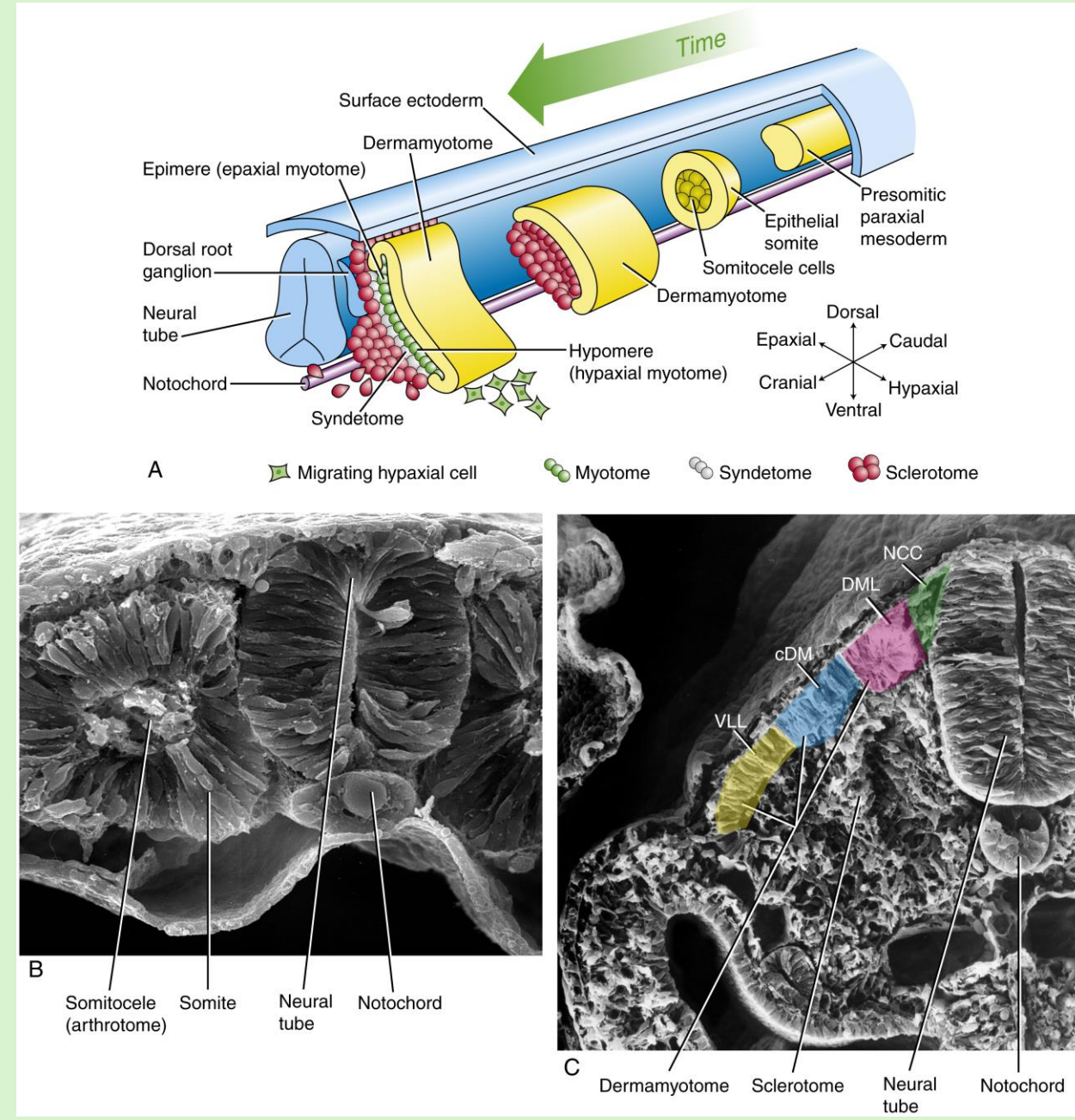
Organogenesis – Weeks 4 to 8

Everything happens synchronously with many simultaneous events

- Formation of the neural tube, spinal cord and nerves, somite differentiation (week 4 & 5)
 - Upper limb buds appear on day 24
 - Lower limb buds appear on day 25/26
 - Development of cartilage model for bone development appear in **week 5**
 - Ossification sites appear in **week 6**
 - Ossification of limb bones occur in **weeks 7 -12**
- 

Recap: Week 3 – Paraxial mesoderm

- The paraxial mesoderm form distinct bodies = **somites**
 - **Epithelial balls with central somitocoele (loosely arranged cells)**
- Identified as bumps on the surface of the developing embryo
- Each somite has three distinct portions
 - **Sclerotome - bones**
 - **Myotome – skeletal muscle**
 - **Dermatome – dermis of skin**
- Each somite is associated with a spinal nerve



General concepts

- Tissue origins
 - Lateral plate mesoderm
 - Bones
 - Tendons
 - Connective tissues
 - Somites
 - Muscles
 - Endothelium

Everything occurs first in the upper limb followed by the same sequence a few days later in the lower

By the end of week 8 they are synchronized

Proximal elements develop first and the distal elements last

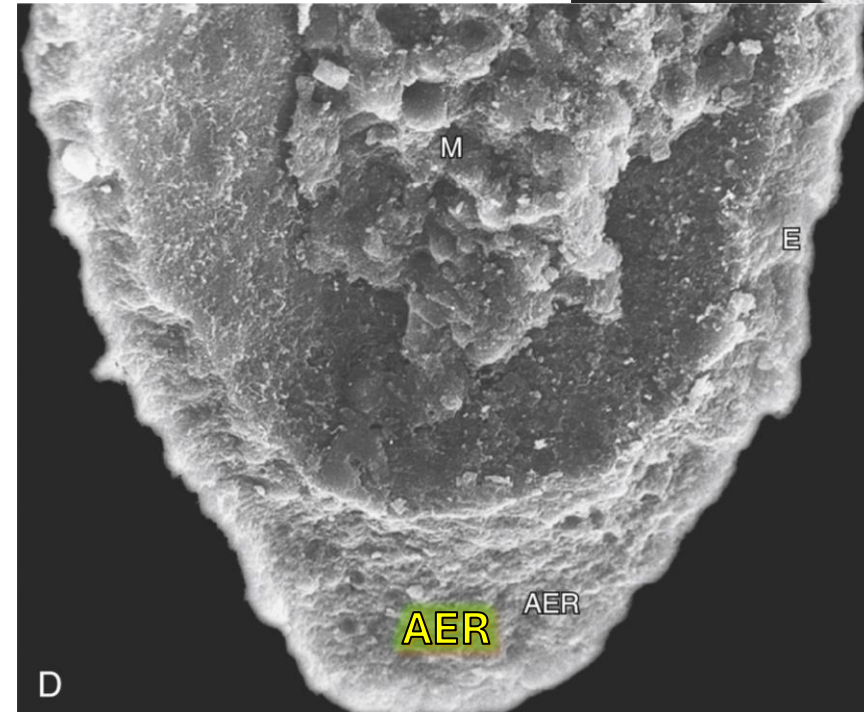
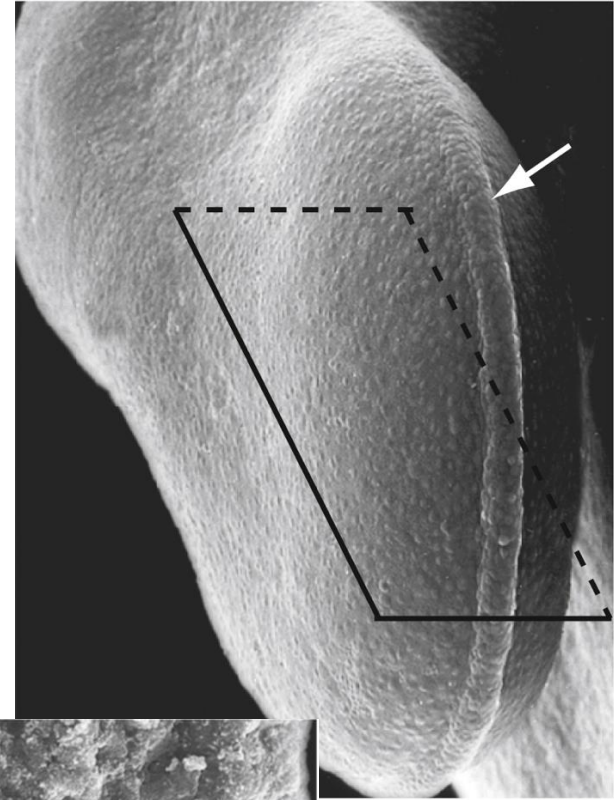
- Patterning
 - Muscles developing on the ventral surface of the long bones
 - Upper limb = flexors & pronators
 - Lower limb = flexors and adductors

Innervated by ventral branches of the ventral primary rami
 - Muscles developing on the dorsal surface of the long bones
 - Upper limb = extensors & supinators
 - Lower limb = extensors & abductors

Innervated by dorsal branches of the ventral primary rami

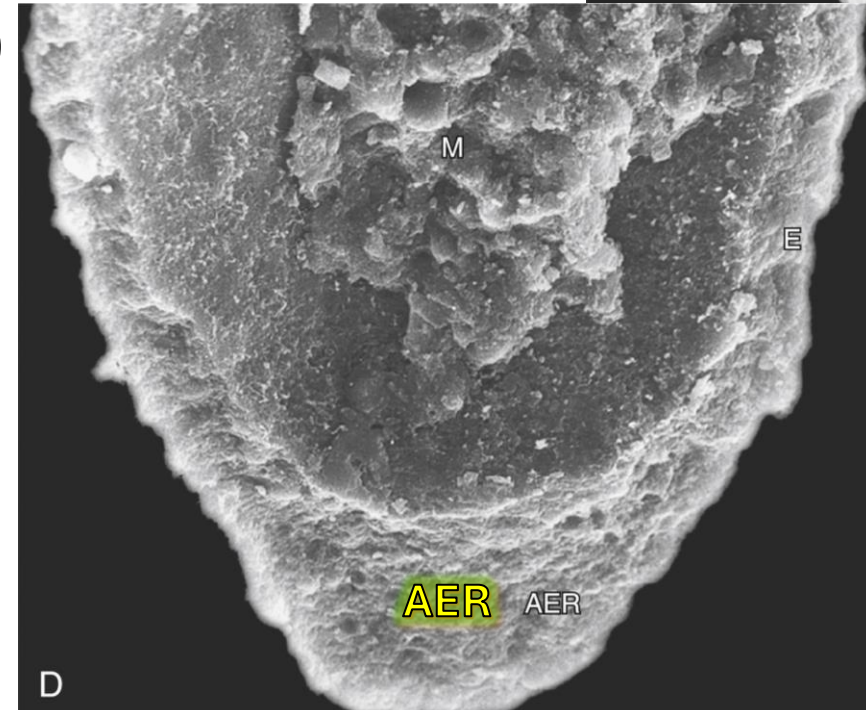
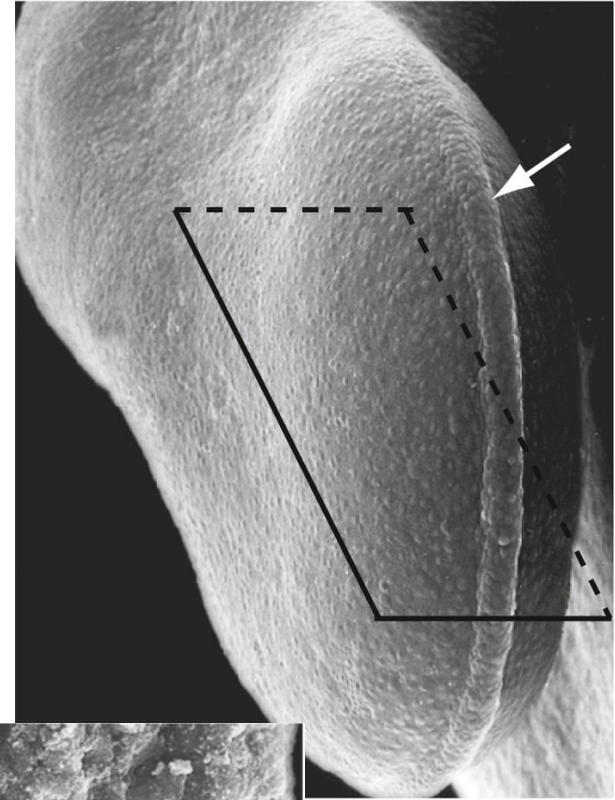
Week 4 – day 24 – Upper limb bud development

- Ventrolateral body wall at the level of **C5 to T1**
- The Homeobox (**HOX**) gene initiates the process
- Activation of mesenchymal cells in the somatic lateral plate mesoderm
- Mesoderm induces formation of a thick band of ectoderm – **Apical Ectodermal Ridge (AER)**
 - **Forms on the ventral-dorsal boundary**
 - Maintains outgrowth of the limb bud along the proximal-distal axis
 - Stimulates proliferation of mesenchyme
- Proliferation and growth are higher on the dorsal side = **development of ventral curvature**

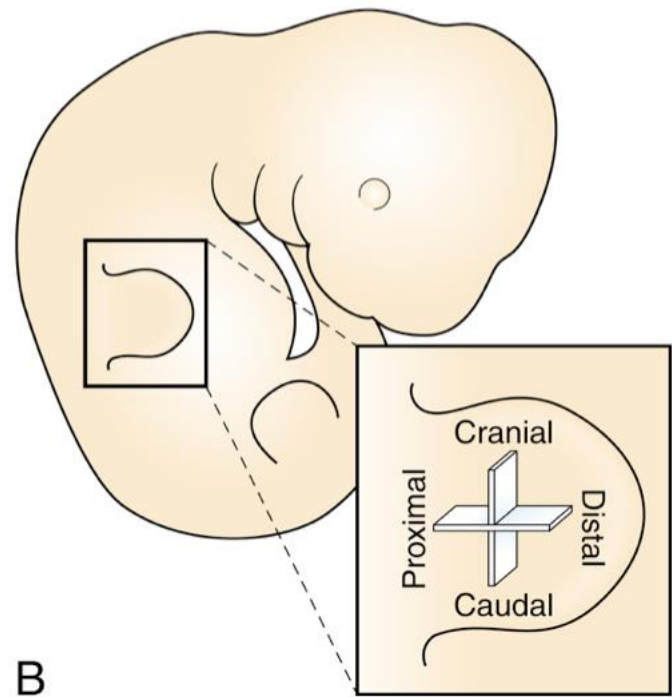


Week 4 – day 28 – Lower limb bud development

- Ventrolateral body wall at the level of **L1 to S1**
- The Homeobox (**HOX**) gene initiates the process
- Activation of mesenchymal cells in the somatic lateral plate mesoderm
- Mesoderm induces formation of a thick band of ectoderm – **Apical Ectodermal Ridge (AER)**
 - **Forms on the ventral-dorsal boundary**
 - Maintains outgrowth of the limb bud along the proximal-distal axis
 - Stimulates proliferation of mesenchyme
- Proliferation and growth are higher on the dorsal side = **development of ventral curvature**



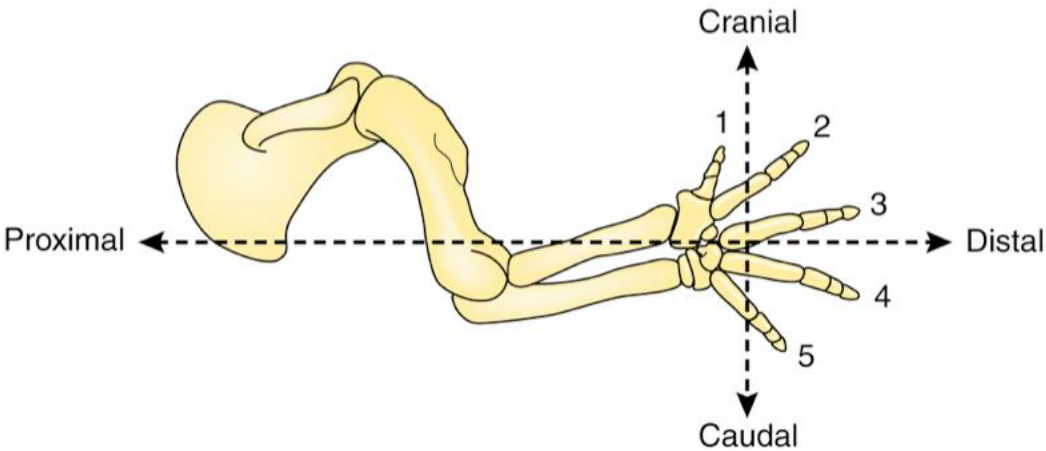
Limb patterning (establishment of axes)



B

Axes and Digits of the Developing Limb

Axis	Signaling Center	Molecular Signal
Proximodistal	Apical ectodermal ridge (AER)	Fibroblast growth factors (FGF-2,-4,-8)
Anteroposterior	Zone of polarizing activity (ZPA)	Sonic hedgehog (Shh)
Dorsoventral	Dorsal ectoderm	Wnt-7a (dorsal)
	Ventral ectoderm	En-1 (ventral)



Week 5

Day 33

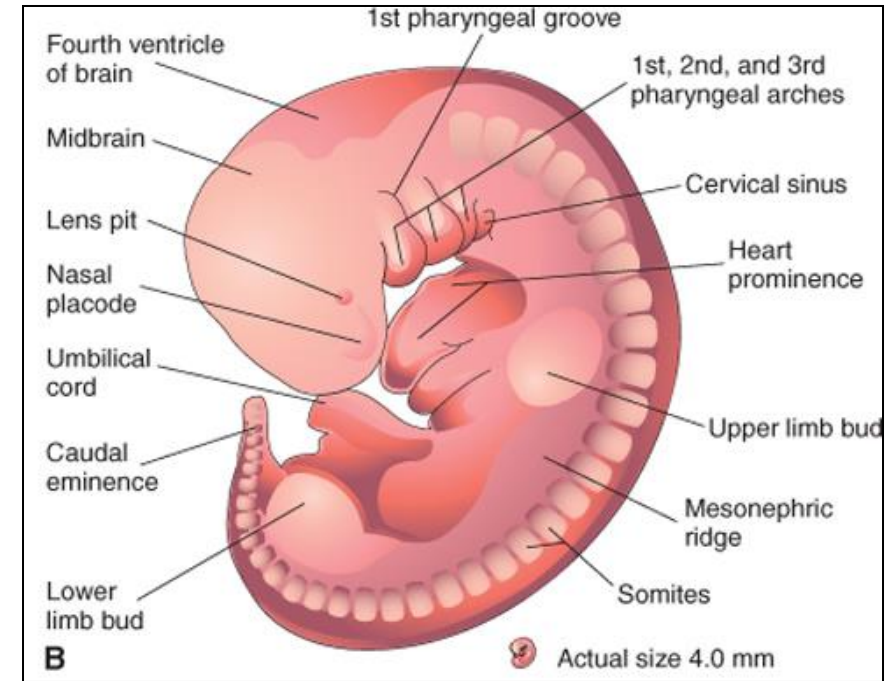


- Upper limb
 - Handplate, forearm, arm and shoulder regions can be distinguished



- Lower limb
 - Rounded proximal part can be distinguished from a more tapering distal part that will form the foot

Mesenchymal condensation giving rise to proximal segments for both limbs is distinct



Week 6

Day 37

- Upper limb
 - Central carpal region surrounded by thinner crescentic rim = **digital plate**
- Lower limb
 - Thigh, leg and foot can be distinguished

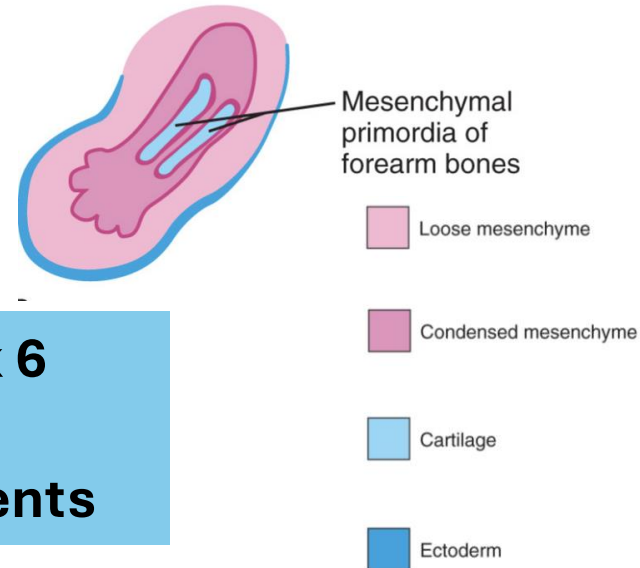
Day 38

- Upper limb
 - Radial thickenings in the digital plate = digital rays
 - **Tips of digital rays project slightly**
- Lower limb
 - Footplate is clearly defined

Upper Limb

38 days

37 days



38 days

37 day

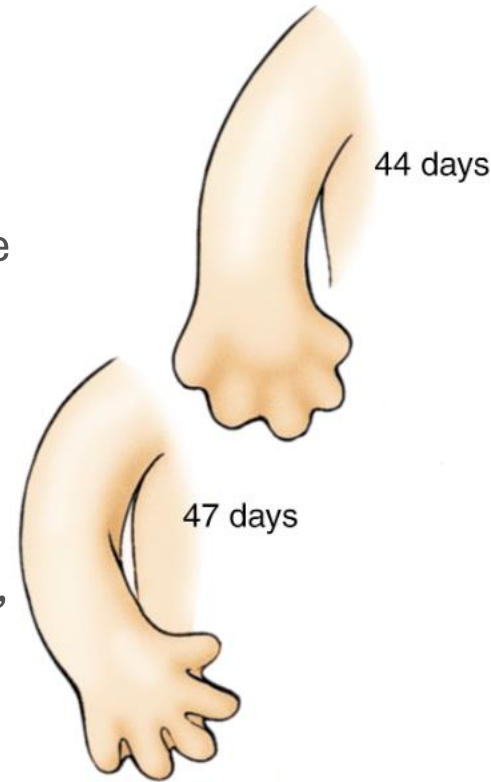
Lower Limb

Week 7

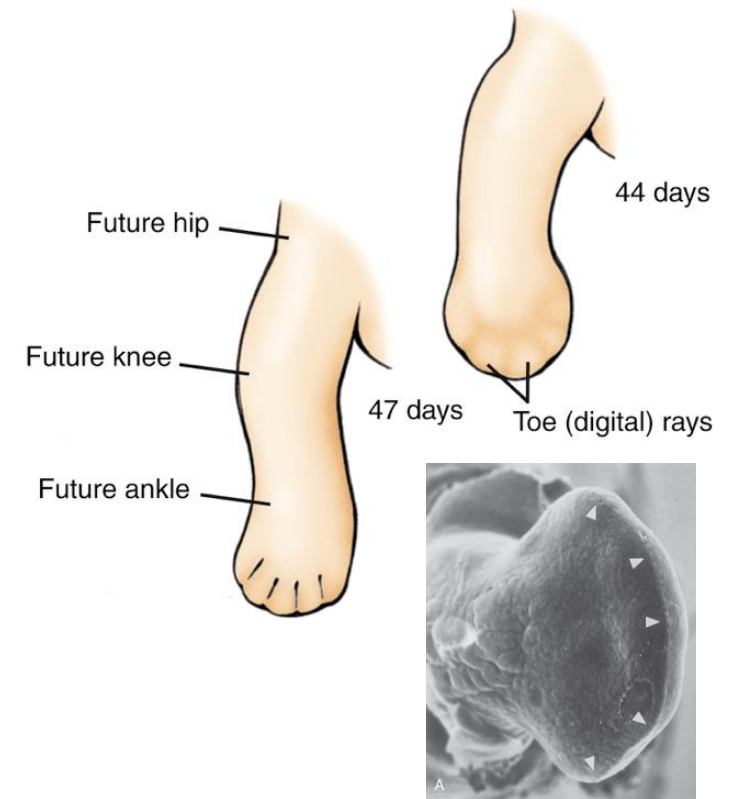
Day 44

- Upper limb
 - Deep notches in distal margin of the digital plate
 - Grooves between the finger rays are deeper
 - Bend where the elbow will form along the proximal-distal axis is becoming defined
- Lower limb
 - Toe rays are visible in the digital plate of the foot, but the rim of the plate is not yet crenulated

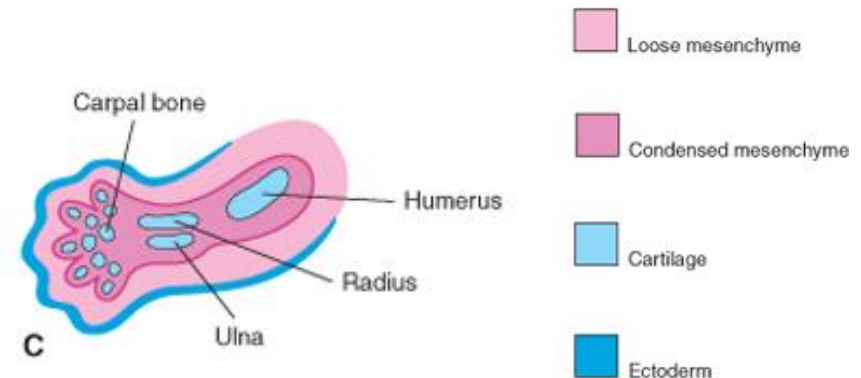
Upper Limb



Lower Limb



Beginning of week 7 (day 41), all skeletal elements of the upper limb except the distal phalanges of 2nd – 5th digits are undergoing chondrification.



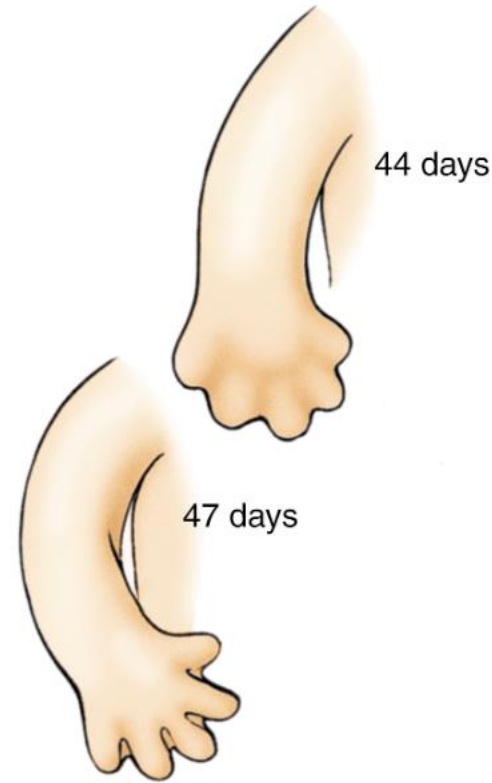
Week 7

Day 47

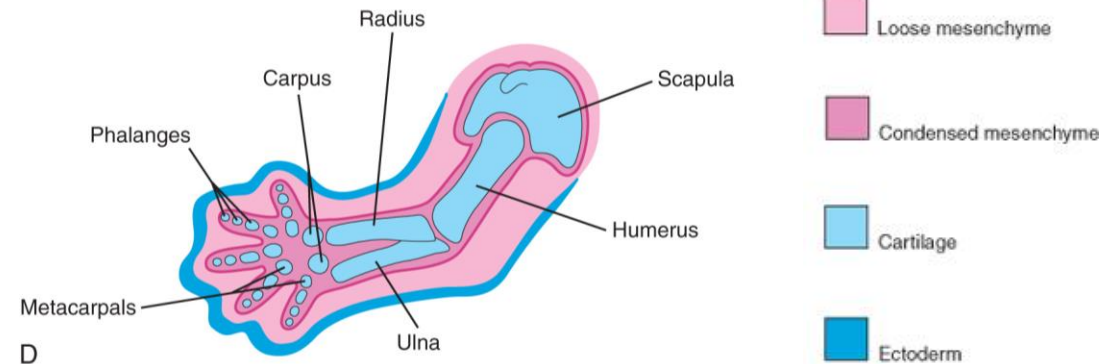
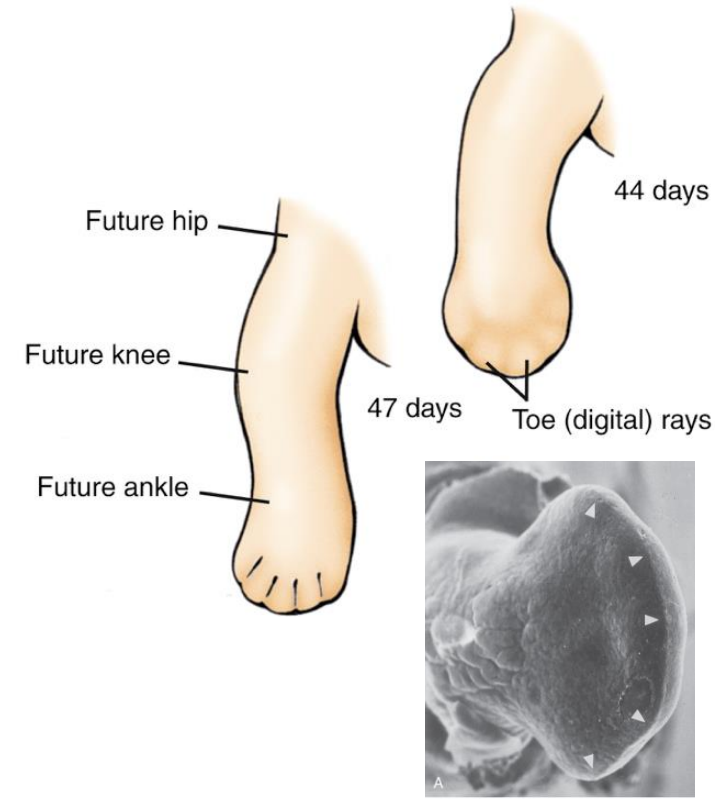
- Upper limb
 - Entire upper limb has undergone ventral flexion
- Lower limb
 - The lower limb has begun to flex toward the midline
 - The toe rays are more prominent, although the margin of the digital plate is still smooth

End of week 7 (day 49), distal phalanges of the hand have begun to chondrify
Chondrification is under way in all elements of the lower limb except the distal row of phalanges

Upper Limb



Lower Limb

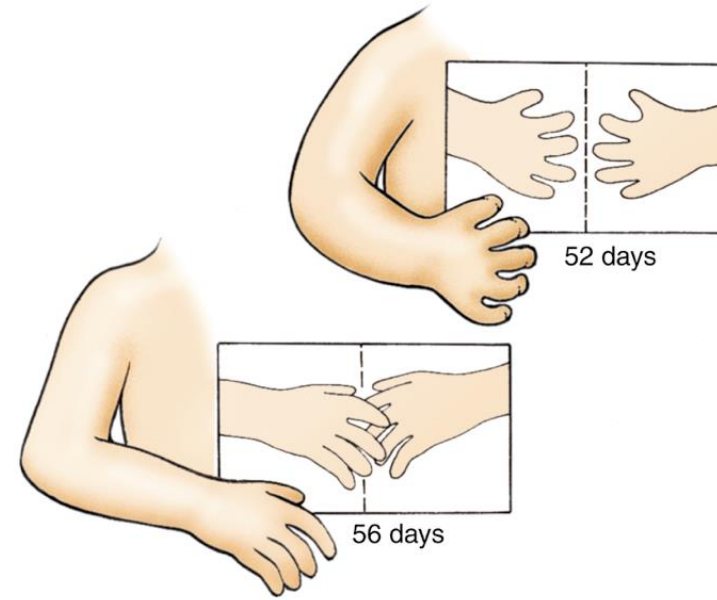


Week 8

Day 52

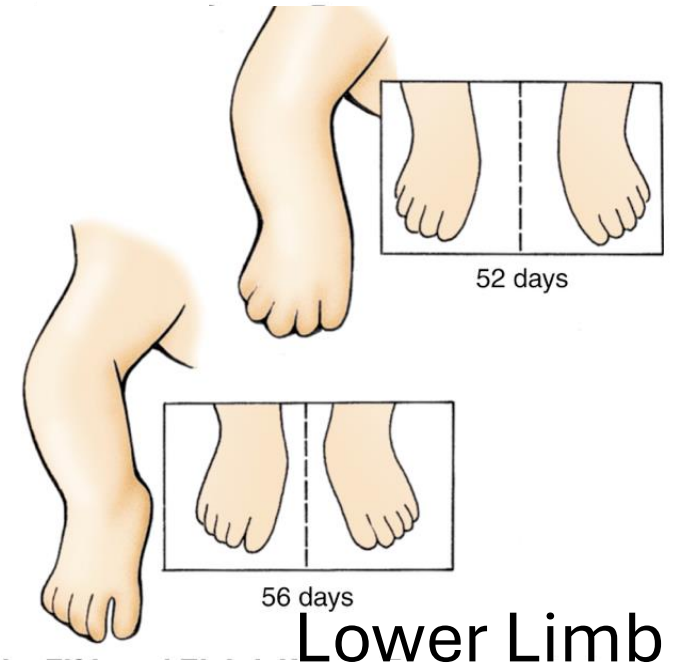
- Upper limb
 - Bent at elbow
 - Distal swellings (tactile pads) on distal fingers
 - Hands flexed at wrist
- Lower limb
 - Longer
 - Rim of the digital plate is notched

Upper Limb



Day 56

- All regions well defined
- Tactile pads on toes visible
- Fingers overlap in the midline



Lower Limb

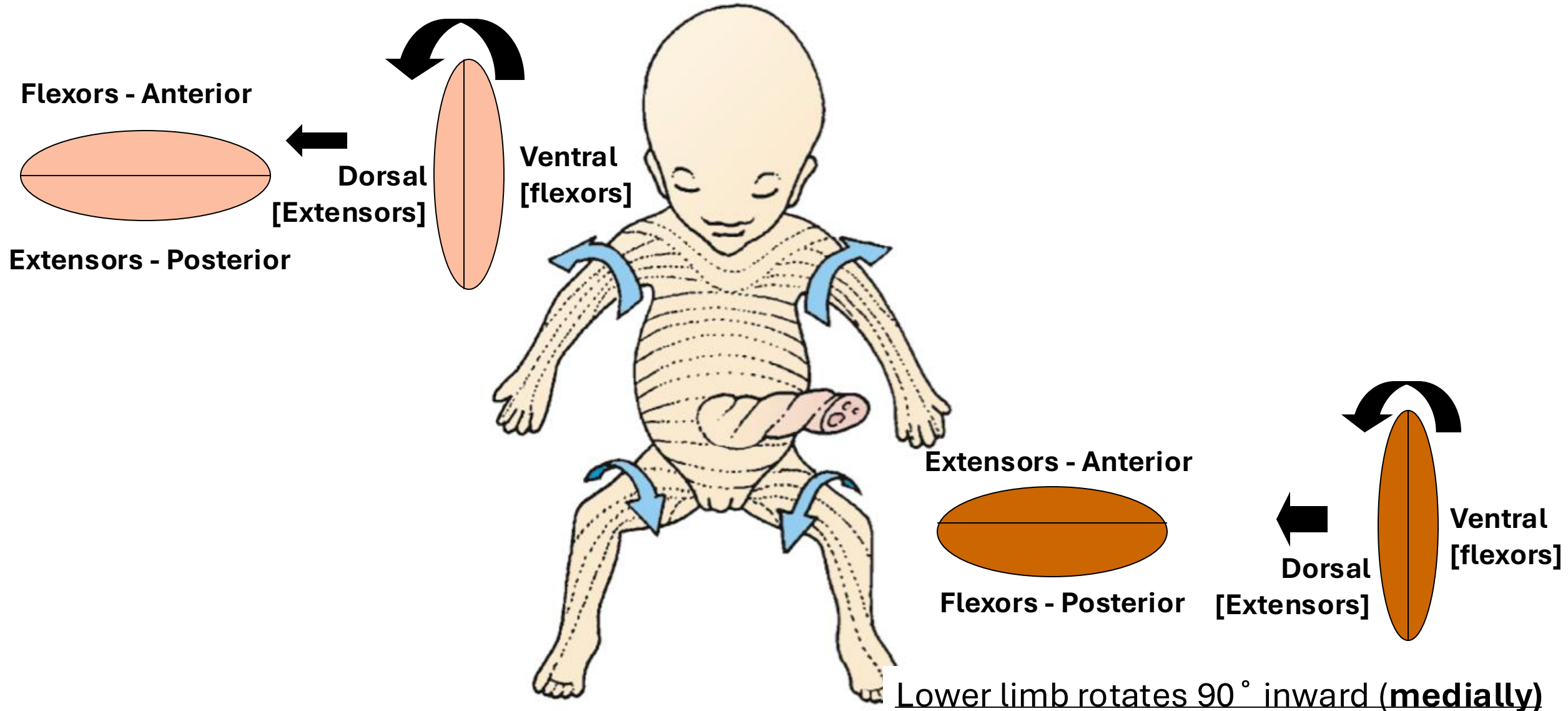
Primary ossification centers of limb bones

- Week 7 – clavicle, humerus, radius and ulna
- Week 8 – Femur and tibia
- Week 9 – Scapula, ilium
- Weeks 11-14 – Metacarpals, metatarsals, distal phalanges, proximal phalanges, middle phalanges
- Week 15 – Ischium
- Week 16 – Calcaneus
- Week 20 – Pubis

- Some smaller bones and sesamoid bones start to ossify after birth – e.g., patella

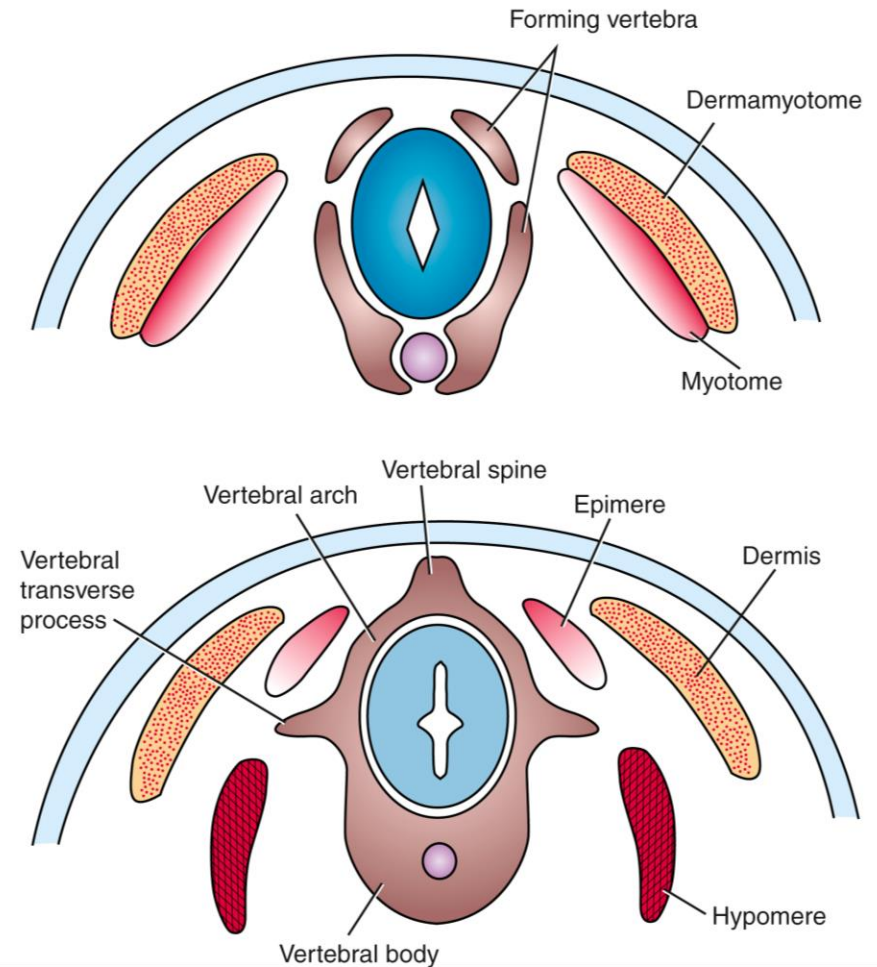
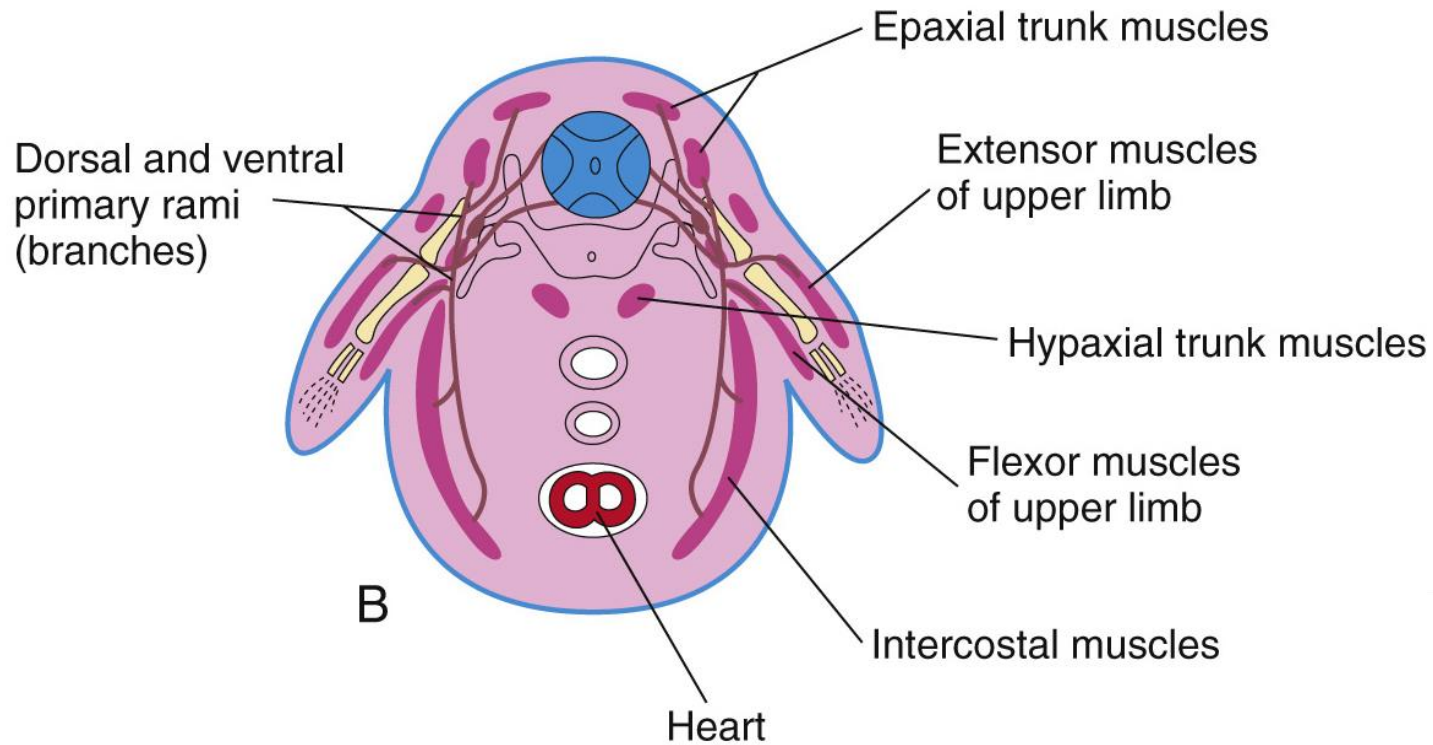
Limb rotation

Upper limb rotates 90° outward (**laterally**)



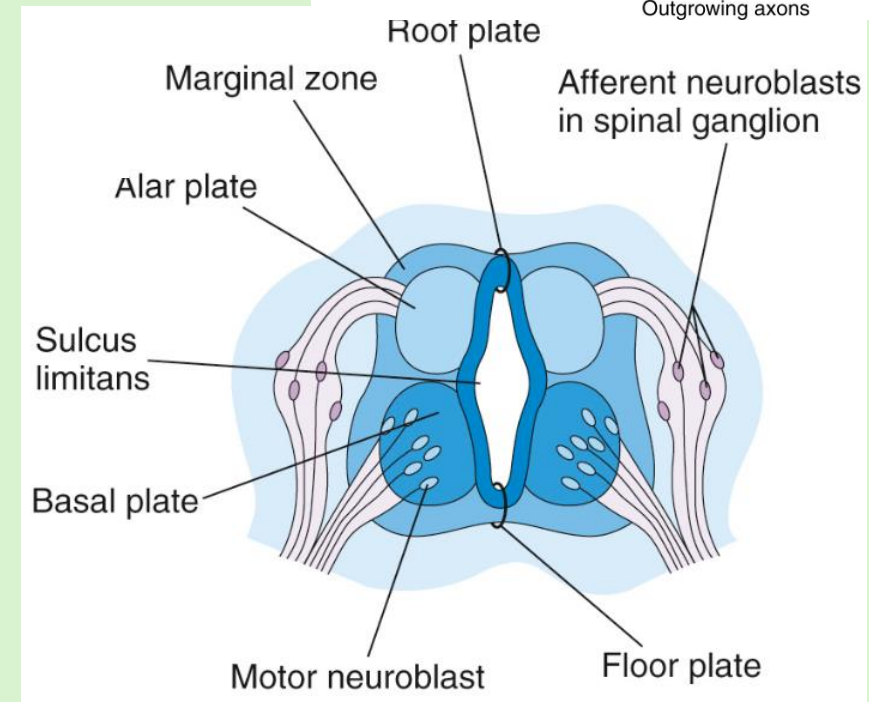
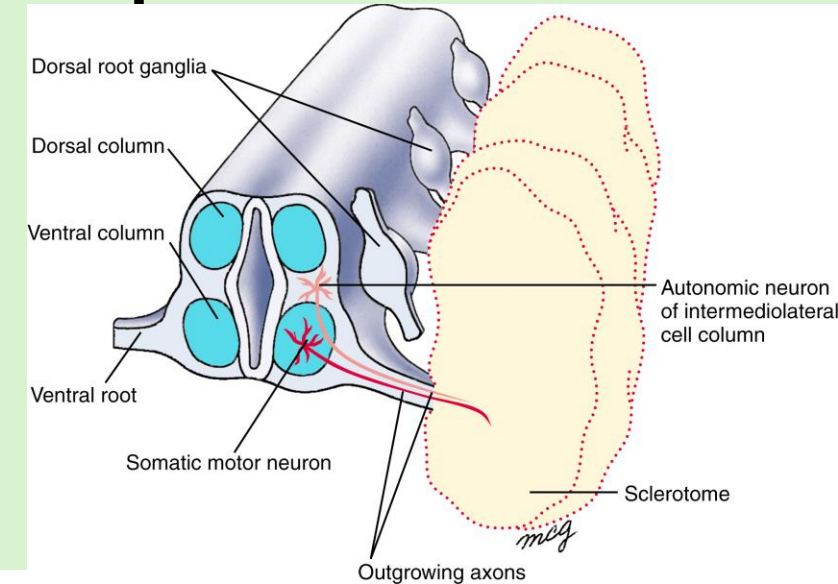
Skeletal muscle development

- Myotome
 - Epimere – epaxial muscles of back
 - Hypomere – **hypaxial** muscles of lateral and ventral body wall and limbs
- Myogenic cells migrate to the developing limb during **week 7**



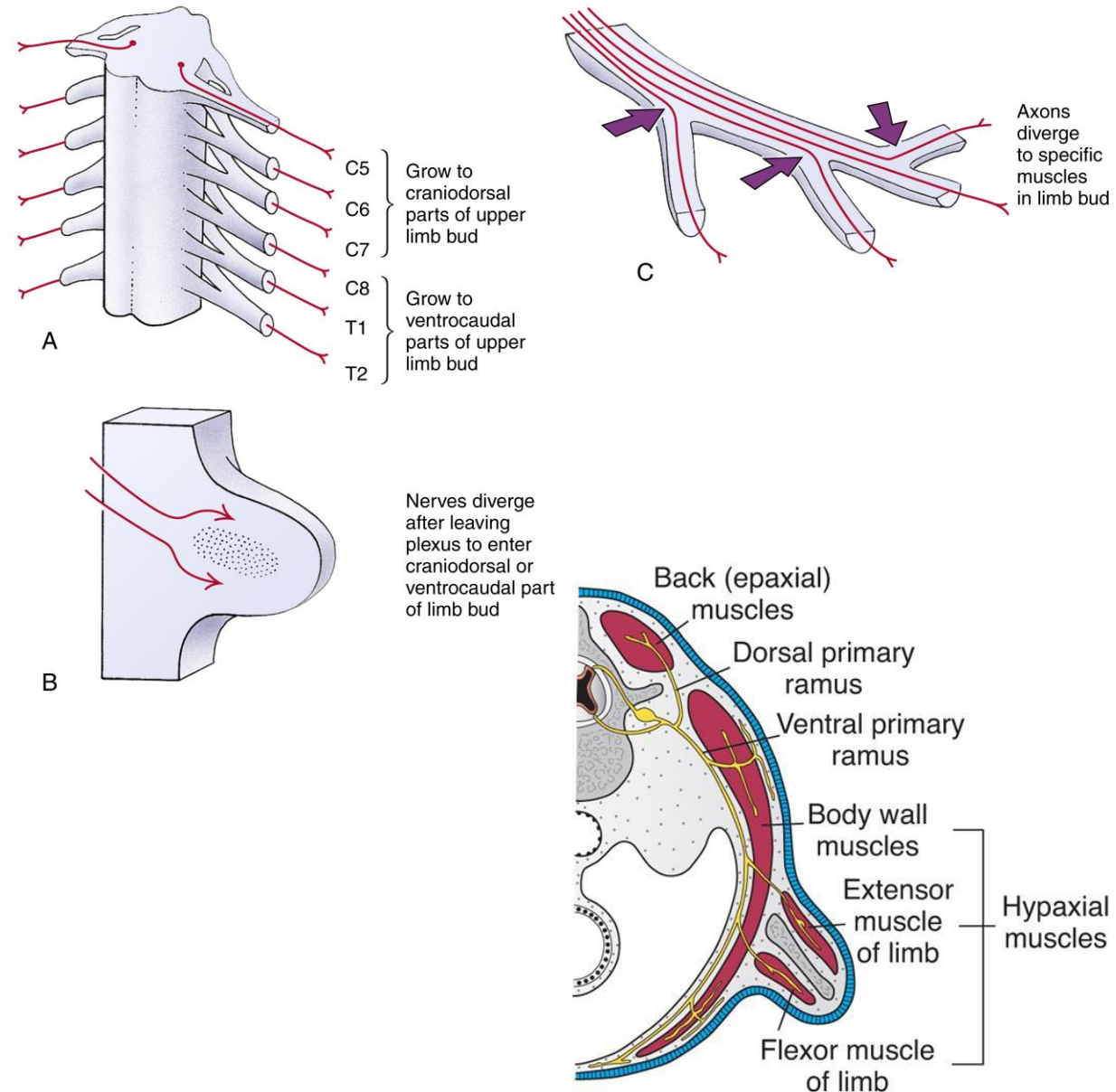
Week 5, day 30 - development of the spinal nerves

- **Axons** from the nerves developing in the **ventral horn** grow toward the **cranial sclerotome**
 - Starts in the neck
 - Initially a continuous broad band
 - Condenses into segmental nerves as they reach the cranial sclerotome
- Neural crest migrates to cranial sclerotome = **dorsal root ganglion**
- **Neurons in the DRG sprout axons in two directions – dorsal roots**
 - Central process synapses with developing neuron in dorsal horn
 - Peripheral process grows toward organs



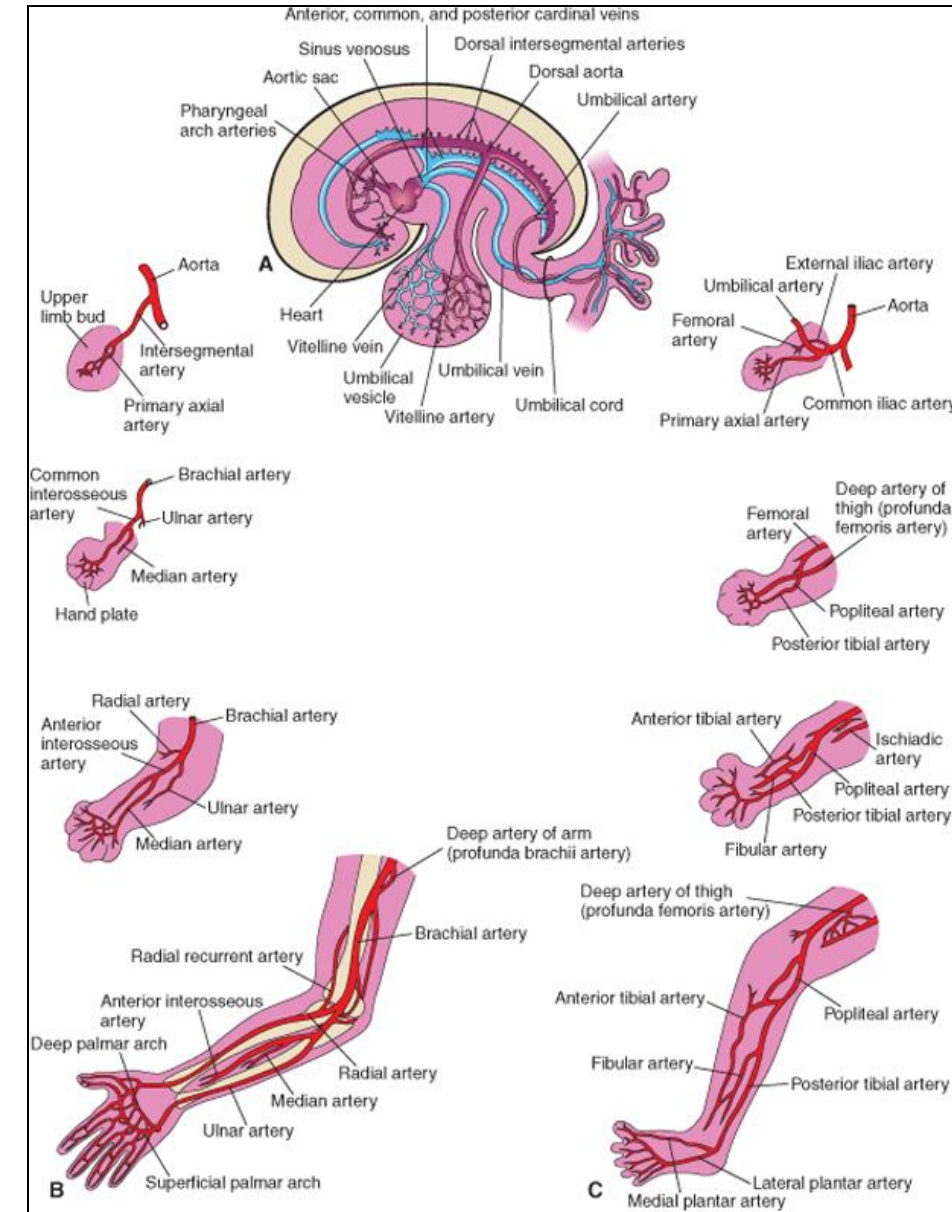
Innervation of skeletal muscles

- **Dorsal rami** innervate the **epaxial muscles**
- **Ventral rami** innervate the **hypaxial muscles**
 - enter the developing limb bud and grow along **permissible pathways**
 - avoid areas of dense mesenchyme or mesenchyme-containing glycosaminoglycans
 - those **arising cranial** to the limb bud will grow toward the **craniodorsal** region
 - those **arising caudal** to limb bud grow toward **ventrocaudal** side
 - once the motor axons arrive at the base of the limb bud, they mix in a specific pattern to form the **plexuses**



Blood vessels of the limbs

- Limb buds supplied by branches of the intersegmental arteries
 - Arise from dorsal aorta
 - Form fine capillary network throughout mesenchyme
- Primordial vascular pattern
 - Primary axial artery & its branches



Summary

- Upper limb develops a few days before the lower limb
- Follow the exact same sequence and patterning
 - Differences lie in rotation
- Develops from lateral plate mesoderm, somitic mesoderm and contains elements from developing nervous system